

ML PROJECT REPORT

School of Technology Management and Engineering

NMIMS, Chandigarh Campus



MACHINE LEARNING

PROJECT REPORT

TOPIC: Predicting audience engagement decline in movies and web series using machine learning

Submitted To:

Ms. Harsimran Kaur

Submitted By:

Ashna Bansal

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1. ABSTRACT

In recent years, streaming platforms have witnessed an exponential increase in the production and release of movies and web series. While some content maintains consistent audience engagement, a significant portion experiences a decline after initial release. This project aims to predict audience engagement decline using machine learning techniques by analyzing various factors such as ratings trends, genre, duration, and content patterns.

The study utilizes datasets collected from publicly available sources and applies data preprocessing, feature engineering, and classification models to identify patterns that contribute to declining engagement. Multiple machine learning models such as Logistic Regression, Random Forest, and Support Vector Machine (SVM) are implemented and evaluated based on performance metrics like accuracy, precision, recall, and F1-score.

The results demonstrate that ensemble-based models like Random Forest perform better in identifying complex patterns in the data. The project provides insights that can help content creators and streaming platforms improve content quality and audience retention strategies.

2. INTRODUCTION

The entertainment industry has undergone a massive transformation with the rise of streaming platforms. Platforms like Netflix, Amazon Prime, and Disney+ release a vast number of movies and web series every year. However, not all content sustains audience interest over time.

Audience engagement decline is a critical issue that affects the success of content. Factors such as weak storytelling, poor pacing, repetitive narratives, or lack of novelty contribute to this decline. Traditionally, such issues are identified only after audience feedback or reduced viewership, which is often too late for corrective measures.

Machine learning provides a powerful solution to this problem by analyzing large datasets and identifying patterns that are not easily visible through manual analysis. This project focuses on using machine learning techniques to predict potential audience engagement decline before it becomes significant.

2.1. BACKGROUND

With the increasing availability of data, the use of machine learning in the entertainment industry has become more prominent. Streaming platforms collect vast amounts of data, including user ratings, watch time, preferences, and interaction patterns.

Analyzing this data can help identify trends and predict outcomes. In the context of movies and web series, machine learning can be used to predict success, popularity, and audience engagement levels.

Previous research has focused primarily on predicting box office success or overall ratings. However, predicting engagement decline is equally important, as it helps in improving content quality and retaining audience interest over time.

2.2. PROBLEM STATEMENT

Movies and web series often experience a decline in audience engagement after their initial release due to various factors such as inconsistent pacing, weak narratives, and repetitive content.

Currently, this decline is identified only after negative feedback or reduced viewership, making it difficult for creators to take preventive action.

The problem addressed in this project is to develop a machine learning model that can predict audience engagement decline using historical data and relevant features, enabling early detection and better decision-making.

2.3. OBJECTIVES

The main objectives of this project are:

- To analyze datasets related to movies and web series
- To identify key factors influencing audience engagement decline
- To preprocess and prepare data for machine learning models
- To apply different machine learning algorithms for prediction
- To evaluate model performance using appropriate metrics
- To provide insights that help improve content strategies

3. LITERATURE REVIEW

Several research studies have explored the use of machine learning in predicting movie success and audience behavior.

- A study on *Movie Success Prediction Using Machine Learning* used classification models such as Logistic Regression, Random Forest, and SVM. The results indicated that Random Forest achieved higher accuracy due to its ability to handle complex data.
- Another research paper titled *Movie Success Prediction System Using Machine Learning* applied Decision Tree, Random Forest, and Gradient Boosting algorithms. It concluded that ensemble methods performed better due to their robustness.
- The study *Movie Box Office Success Prediction* compared Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree models, where Decision Tree and KNN showed better performance.
- A more advanced approach involved ensemble learning models combining multiple algorithms, which significantly improved prediction accuracy.
- Research on early prediction emphasized the importance of identifying trends before release or during early stages using Support Vector Machines and Random Forest.

Overall, the literature suggests that ensemble models and advanced feature engineering play a crucial role in improving prediction performance.

4. METHODOLOGY

The methodology of this project involves several stages, starting from data collection to model evaluation.

4.1. DATA COLLECTION

The dataset used in this project was collected from publicly available sources such as Kaggle. It includes information about movies and web series, such as:

- Ratings
- Genre
- Duration
- Release details
- Popularity metrics

Multiple datasets were combined to enrich the data and improve model performance.

Dataset 1:

```
In [16]: df1=pd.read_csv("D:/SEMA/Audience engagement ML Project/Datasets/archive (1)/netflix_titles Nov 2019.csv")
df1.head()
```

Out[16]:	show_id	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	type
0	81193313	Chocolate	NaN	Ha Ji-won, Yoon Kye-sang, Jang Seung-jo, Kang ...	South Korea	November 30, 2019	2019	TV-14	1 Season	International TV Shows, Korean TV Shows, Roman...	Brought together by meaningful meals in the pa...	TV Show
1	81197050	Guatemala: Heart of the Mayan World	Luis Ara, Ignacio Jaunsolo	Christian Morales	NaN	November 30, 2019	2019	TV-G	67 min	Documentaries, International Movies	From Sierra de las Minas to Esquipulas, explor...	Movie
2	81213894	The Zoya Factor	Abhishek Sharma	Sonam Kapoor, Dulquer Salmaan, Sanjay Kapoor, ...	India	November 30, 2019	2019	TV-14	135 min	Comedies, Dramas, International Movies	A goofy copywriter unwittingly convinces the L...	Movie
3	81082007	Atlantics	Mati Diop	Mama Sane, Amadou Mbow, Ibrahima Traore, Nicol...	France, Senegal, Belgium	November 29, 2019	2019	TV-14	106 min	Dramas, Independent Movies, International Movies	Arranged to marry a rich man, young Ada is cru...	Movie
4	80213643	Chip and Potato	NaN	Abigail Oliver, Andrea Libman, Briana Buckmast...	Canada, United Kingdom	NaN	2019	TV-Y	2 Seasons	Kids' TV	Lovable pug Chip starts kindergarten, makes ne...	TV Show

Dataset 2:


```
In [17]: df2=pd.read_csv("D:/SEM4/Audience engagement ML Project/Datasets/archive (2)/imdb_top_1000.csv")
df2.head()
```

```
Out[17]:
```

	Poster_Link	Series_Title	Released_Year	Certificate	Runtime	Genre	IMDB_Rating	Overview	Meta_score	Director	Star1	Star2	Star3	Star4	No_of_Votes	Gross
0	https://m.media-amazon.com/images/M/MV5BMDFkYT...	The Shawshank Redemption	1994	A	142 min	Drama	9.3	Two imprisoned men bond over a number of years...	80.0	Frank Darabont	Tim Robbins	Morgan Freeman	Bob Gunton	William Sadler	2343110	28,341,469
1	https://m.media-amazon.com/images/M/MV5BM2MyNj...	The Godfather	1972	A	175 min	Crime, Drama	9.2	An organized crime dynasty's aging patriarch L...	100.0	Francis Ford Coppola	Marlon Brando	Al Pacino	James Caan	Diane Keaton	1620367	134,966,411
2	https://m.media-amazon.com/images/M/MV5BMTMxNT...	The Dark Knight	2008	UA	152 min	Action, Crime, Drama	9.0	When the menace known as the Joker wreaks havoc...	84.0	Christopher Nolan	Christian Bale	Heath Ledger	Aaron Eckhart	Michael Caine	2303232	534,858,444
3	https://m.media-amazon.com/images/M/MV5BMWwMG...	The Godfather: Part II	1974	A	202 min	Crime, Drama	9.0	The early life and career of Vito Corleone in ...	90.0	Francis Ford Coppola	Al Pacino	Robert De Niro	Robert Duval	Diane Keaton	1129952	57,300,000
4	https://m.media-amazon.com/images/M/MV5BMWU4N2...	12 Angry Men	1957	U	96 min	Crime, Drama	9.0	A jury holdout attempts to prevent a miscarria...	96.0	Sidney Lumet	Henry Fonda	Lee J. Cobb	Martin Balsam	John Fiedler	689845	4,360,000

Dataset 3:

```
In [18]: df3=pd.read_csv("D:/SEM4/Audience engagement ML Project/Datasets/archive (3)/tmdb_5000_movies.csv")
df3.head()
```

```
Out[18]:
```

	budget	genres	homepage	id	keywords	original_language	original_title	overview	popularity	production_companies	production_countries	release_date	revenue	runtime	spoken_languages	status	tagline	title	vote_average	vote_count
0	237000000	["id": 28, "name": "Action"], ["id": 12, "name": ...]	http://www.avatarmovie.com/	19995	["id": 1463, "name": "culture clash"], ["id": ...]	en	Avatar	In the 22nd century, a paraplegic Marine is di...	150.437577	["name": "Ingenious Film Partners", "id": 289...]	["iso_3166_1": "US", "name": "United States o...]	2009-12-10	2787965307	162.0	["iso_639_1": "en", "name": "English"], ["iso_639_1": "en", "name": "English"]	Released	Enter the World of Pandora.	Avatar	7.2	11800
1	300000000	["id": 12, "name": "Adventure"], ["id": 14, "name": ...]	http://disney.go.com/disneypictures/pirates/	285	["id": 270, "name": "ocean"], ["id": 726, "name": "ra...]	en	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	139.082615	["name": "Walt Disney Pictures", "id": 2], [...]	["iso_3166_1": "US", "name": "United States o...]	2007-05-19	961000000	169.0	["iso_639_1": "en", "name": "English"], ["iso_639_1": "en", "name": "English"]	Released	At the end of the world, the Caribbean At adventure begins.	Pirates of the Caribbean: At World's End	6.9	4300
2	245000000	["id": 28, "name": "Action"], ["id": 12, "name": ...]	https://www.sonypictures.com/movies/spectre/	206647	["id": 470, "name": "spy"], ["id": 818, "name": ...]	en	Spectre	A cryptic message from Bond's past sends him o...	107.376788	["name": "Columbia Pictures", "id": 5], ["name": ...]	["iso_3166_1": "GB", "name": "United Kingdom"], ["iso_639_1": "fr", "name": "Fran\u00e7ais"], ["iso_639_1": "en", "name": "English"]	2015-10-26	880674639	148.0	["iso_639_1": "fr", "name": "Fran\u00e7ais"], ["iso_639_1": "en", "name": "English"]	Released	A Plan No One Escapes	Spectre	6.3	4466
3	250000000	["id": 28, "name": "Action"], ["id": 80, "name": ...]	http://www.thedarkknightrises.com/	49026	["id": 848, "name": "dc comics"], ["id": 653, "name": ...]	en	The Dark Knight Rises	Following the death of District Attorney Hane...	112.312950	["name": "Legendary Pictures", "id": 923], [...]	["iso_3166_1": "US", "name": "United States o...]	2012-07-16	1084939999	165.0	["iso_639_1": "en", "name": "English"], ["iso_639_1": "en", "name": "English"]	Released	The Legend Ends	The Dark Knight Rises	7.6	9106
4	260000000	["id": 28, "name": "Action"], ["id": 12, "name": ...]	http://movies.disney.com/john-carter	49529	["id": 818, "name": "based on novel"], ["id": ...]	en	John Carter	John Carter is a war-weary, former military ca...	43.826995	["name": "Walt Disney Pictures", "id": 2], ["name": ...]	["iso_3166_1": "US", "name": "United States o...]	2012-03-07	284139100	132.0	["iso_639_1": "en", "name": "English"], ["iso_639_1": "en", "name": "English"]	Released	Lost in our world, found in another.	John Carter	6.1	2124

Dataset 4:

```
In [19]: df4=pd.read_csv("D:/SEM4/Audience engagement ML Project/Datasets/archive (3)/tmdb_5000_credits.csv")
df4.head()
```

```
Out[19]:
```

	movie_id	title	cast	crew
0	19995	Avatar	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...
1	285	Pirates of the Caribbean: At World's End	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit_id": "52fe4232c3a36847f800b579", "de...
2	206647	Spectre	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit_id": "54805967c3a36829b5002c41", "de...
3	49026	The Dark Knight Rises	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"credit_id": "52fe4781c3a36847f81398c3", "de...
4	49529	John Carter	[{"cast_id": 5, "character": "John Carter", "c...	[{"credit_id": "52fe479ac3a36847f81398c3", "de...

Merged dataset:

[17]:	budget	genres	homepage	id	keywords	original_language	original_title	overview	popularity	production_compa
0	237000000	{{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	{{"id": 1463, "name": "culture clash"}, {"id":...	en	Avatar	In the 22nd century, a paraplegic Marine is di...	150.437577	{{"name": "Ingeni Film Partners", "id"
1	300000000	{{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	{{"id": 270, "name": "ocean"}, {"id": 726, "na...	en	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	139.082615	{{"name": "Walt Dis Pictures", "id": 2),
2	245000000	{{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.sonypictures.com/movies/spectre/	206647	{{"id": 470, "name": "spy"}, {"id": 818, "name...	en	Spectre	A cryptic message from Bond's past sends him o...	107.376788	{{"name": "Colum Pictures", "id" ("na
3	250000000	{{"id": 28, "name": "Action"}, {"id": 80, "nam...	http://www.thedarkknightrises.com/	49026	{{"id": 849, "name": "dc comics"}, {"id": 853,...	en	The Dark Knight Rises	Following the death of District Attorney Harve...	112.312950	{{"name": "Legend Pictures", "id": 923),
4	260000000	{{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://movies.disney.com/john-carter	49529	{{"id": 818, "name": "based on novel"}, {"id":...	en	John Carter	John Carter is a war- weary, former military ca...	43.926995	{{"name": "Walt Dis Pictures", "id"

5 rows x 24 columns

4.2. DATA PREPROCESSING

Data preprocessing is a crucial step in machine learning. The following steps were performed:

- Understanding the data

```
[344]: df.describe()
```

	budget	popularity	revenue	runtime	vote_average	vote_count	success	predicted_success	engagement_score
count	4.803000e+03	4803.000000	4.803000e+03	4803.000000	4803.000000	4803.000000	4803.000000	4803.000000	4803.000000
mean	2.904504e+07	21.492301	8.226064e+07	106.875859	6.092172	690.217989	0.499896	0.505309	351.235106
std	4.072239e+07	31.816650	1.628571e+08	22.607225	1.194612	1234.585891	0.500052	0.500024	622.369654
min	0.000000e+00	0.000000	0.000000e+00	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	7.900000e+05	4.668070	0.000000e+00	94.000000	5.600000	54.000000	0.000000	0.000000	29.644895
50%	1.500000e+07	12.921594	1.917000e+07	103.000000	6.200000	235.000000	0.000000	1.000000	122.413050
75%	4.000000e+07	28.313505	9.291719e+07	117.500000	6.800000	737.000000	1.000000	1.000000	376.454887
max	3.800000e+08	875.581305	2.787965e+09	338.000000	10.000000	13752.000000	1.000000	1.000000	6911.946742

- Handling missing values

```
In [266]: print("Shape:", df.shape)
print("Columns:", df.columns)
print("Null values after merge:", df.isnull().sum())

Shape: (4803, 24)
Columns: Index(['budget', 'genres', 'homepage', 'id', 'keywords', 'original_language',
               'original_title', 'overview', 'popularity', 'production_companies',
               'production_countries', 'release_date', 'revenue', 'runtime',
               'spoken_languages', 'status', 'tagline', 'title', 'vote_average',
               'vote_count', 'movie_id', 'title', 'cast', 'crew'],
              dtype='object')
Null values after merge: budget      0
genres      0
homepage    3091
id          0
keywords    0
original_language  0
original_title  0
overview    3
popularity  0
production_companies  0
production_countries  0
release_date  1
revenue     0
runtime     2
spoken_languages  0
status      0
tagline     844
title       0
vote_average  0
vote_count  0
movie_id    0
title       0
cast        0
crew        0
dtype: int64
```

```
In [269]: df = df.drop(['homepage', 'tagline'], axis=1)
```

- Removing duplicate entries

```
[213]: #cleaning
duplicate_count = df.duplicated().sum()
print("Duplicate rows:", duplicate_count)

Duplicate rows: 0

[214]: df[df.duplicated()]

[214]: budget genres homepage id keywords original_language original_title overview popularity production_companies ... spoken_languages status tagline tit

0 rows x 24 columns
```



```
[215]: df = df.drop(['homepage', 'tagline'], axis=1)
```

- Binary classification

```
[225]: #binary classification
df['success'] = df['revenue'].apply(lambda x: 1 if x > df['revenue'].median() else 0)

[226]: df
```

- Normalizing numerical values

```
from sklearn.feature_extraction.text import CountVectorizer

cv = CountVectorizer(max_features=5000)
text_features = cv.fit_transform(df['combined_features']).toarray()

print(text_features.shape)

(4803, 5000)
```

- Cleaning inconsistent data

```
[217]: df.shape

[217]: (4803, 22)

[218]: df.columns[df.columns.duplicated()]

[218]: Index(['title'], dtype='object')

[219]: df = df.loc[:, ~df.columns.duplicated()]
```

This ensured that the dataset was accurate and suitable for model training.

4.3. FEATURE SELECTION/ENGINEERING

Feature engineering involved selecting and creating relevant features that influence audience engagement.

- Combining multiple attributes (e.g., genre + ratings trends)

```
[229]: #feature selection
df = df.drop(columns=[
    'id',
    'movie_id',
    'overview',
    'production_companies',
    'production_countries',
    'spoken_languages',
    'status',
    'original_language',
    'release_date'
])
```

- Converting JSON data into usable format

```
[48]: import ast

[49]: def extract_names(text):
    names = []
    try:
        for item in ast.literal_eval(text):
            names.append(item['name'])
    except:
        return ''
    return ' '.join(names)

[50]: df['genres'] = df['genres'].apply(extract_names)
df['keywords'] = df['keywords'].apply(extract_names)
```

```
[51]: def extract_cast(text):
    names = []
    try:
        for item in ast.literal_eval(text)[:3]:
            names.append(item['name'])
    except:
        return ''
    return ' '.join(names)

df['cast'] = df['cast'].apply(extract_cast)
```

```
[52]: def extract_director(text):
    names = []
    try:
        for item in ast.literal_eval(text):
            if item['job'] == 'Director':
                names.append(item['name'])
    except:
        return ''
    return ' '.join(names)

df['crew'] = df['crew'].apply(extract_director)
```

```
[53]: print(df['genres'].head())
print(df['keywords'].head())
print(df['cast'].head())
print(df['crew'].head())

0    Action Adventure Fantasy Science Fiction
1             Adventure Fantasy Action
2             Action Adventure Crime
3             Action Crime Drama Thriller
4    Action Adventure Science Fiction
Name: genres, dtype: object
0    culture clash future space war space colony so...
1    ocean drug abuse exotic island east india trad...
2    spy based on novel secret agent sequel mi6 bri...
3    dc comics crime fighter terrorist secret ident...
4    based on novel mars medallion space travel pri...
Name: keywords, dtype: object
0    Sam Worthington Zoe Saldana Sigourney Weaver
1    Johnny Depp Orlando Bloom Keira Knightley
2    Daniel Craig Christoph Waltz Léa Seydoux
3    Christian Bale Michael Caine Gary Oldman
4    Taylor Kitsch Lynn Collins Samantha Morton
Name: cast, dtype: object
0    James Cameron
1    Gore Verbinski
2    Sam Mendes
3    Christopher Nolan
4    Andrew Stanton
Name: crew, dtype: object
```

- Creating new features based on engagement patterns

```
[90]: df['engagement_score'] = (
        df['vote_count'] * 0.5 +
        df['vote_average'] * 0.3 +
        df['popularity'] * 0.2
    )

[92]: df['engagement_level'] = df['engagement_score'].apply(
        lambda x: 'High' if x > df['engagement_score'].mean() else 'Low'
    )
```

Proper feature selection helps improve model accuracy and reduces overfitting.

4.4. MODEL SELECTION

Model Train – Test split:

```
[64]: #model traintest split
import numpy as np

numeric_features = df[['budget', 'popularity', 'runtime', 'vote_average', 'vote_count']].values

X = np.hstack((numeric_features, text_features))
y = df['success']

[65]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

The following machine learning models were used:

- Logistic Regression

```
[66]: #training of models
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

[67]: #lr
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)

acc_lr = accuracy_score(y_test, y_pred_lr)
print("Logistic Regression Accuracy:", acc_lr)

Logistic Regression Accuracy: 0.8553590010405827
```

- KNN

```
[68]: #knn
      knn = KNeighborsClassifier(n_neighbors=5)
      knn.fit(X_train, y_train)
      y_pred_knn = knn.predict(X_test)

      acc_knn = accuracy_score(y_test, y_pred_knn)
      print("KNN Accuracy:", acc_knn)

      KNN Accuracy: 0.8459937565036421
```

- Random Forest

```
[69]: #random forest
      rf = RandomForestClassifier()
      rf.fit(X_train, y_train)
      y_pred_rf = rf.predict(X_test)

      acc_rf = accuracy_score(y_test, y_pred_rf)
      print("Random Forest Accuracy:", acc_rf)

      Random Forest Accuracy: 0.8616024973985432
```

- Support Vector Machine (SVM)

```
[70]: #svm
      svm = SVC()
      svm.fit(X_train, y_train)
      y_pred_svm = svm.predict(X_test)

      acc_svm = accuracy_score(y_test, y_pred_svm)
      print("SVM Accuracy:", acc_svm)

      SVM Accuracy: 0.809573361082206
```

- Gradient Boosting

```
[71]: #gradient boosting
      gb = GradientBoostingClassifier()
      gb.fit(X_train, y_train)
      y_pred_gb = gb.predict(X_test)

      acc_gb = accuracy_score(y_test, y_pred_gb)
      print("Gradient Boosting Accuracy:", acc_gb)

      Gradient Boosting Accuracy: 0.8668054110301769
```

These models were selected based on their effectiveness in classification problems.

4.5. TOOLS & TECHNOLOGIES USED

The project was implemented using the following tools:

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Jupyter Notebook

These tools helped in data processing, visualization, and model implementation.

5. IMPLEMENTATION

The implementation phase involved several structured steps to ensure accurate prediction of audience engagement decline:

- Merging multiple datasets:

Data from different sources was combined to create a comprehensive dataset containing movie attributes such as ratings, genres, duration, and popularity metrics.

- Cleaning and preprocessing data:

Missing values were handled, duplicate records were removed, and inconsistent data formats were corrected. Numerical features were scaled where necessary to improve model performance.

- Performing exploratory data analysis (EDA):

Visualizations and statistical analysis were used to understand data distribution, identify patterns, and detect correlations between features affecting audience engagement.

- Splitting data into training and testing sets:

The dataset was divided into training and testing subsets (typically 80:20 ratio) to evaluate the model's ability to generalize to unseen data.

- Training machine learning models:

Various models including Logistic Regression, Random Forest, SVM, KNN, and Gradient Boosting were trained using the training dataset.

- Evaluating model performance:

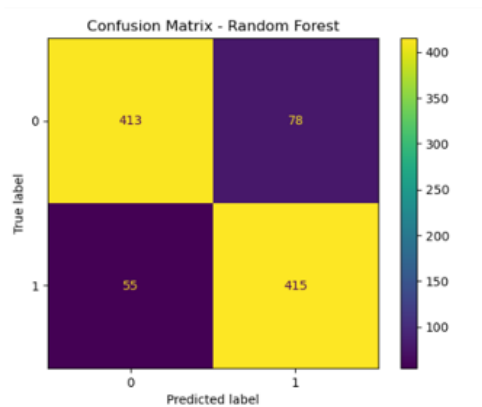
The trained models were tested using the test dataset and evaluated using metrics such as accuracy, precision, recall, and F1-score.

6. RESULTS AND EVALUATION

After training the selected machine learning models, their performance was evaluated using standard classification metrics. This section presents the results obtained and compares the effectiveness of each model in predicting audience engagement decline.

6.1. PERFORMANCE METRICS

Confusion matrix was also used to visualize model performance.



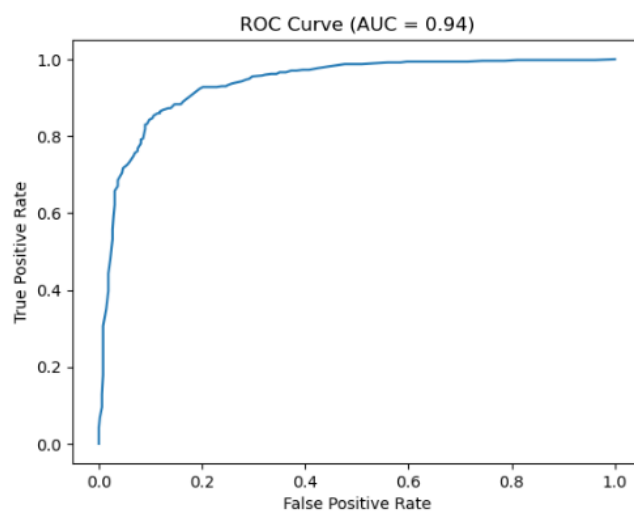
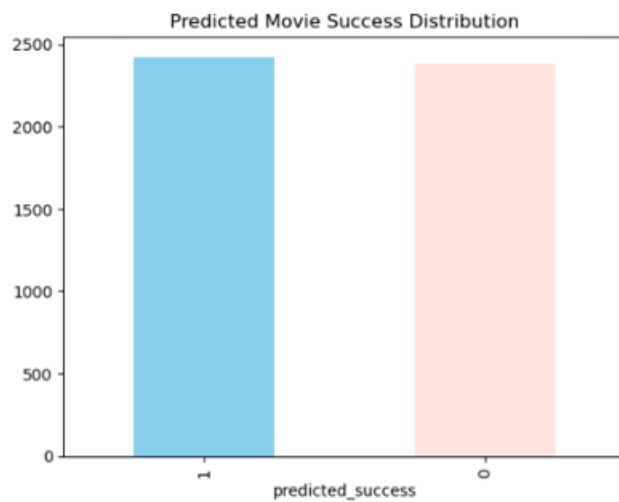
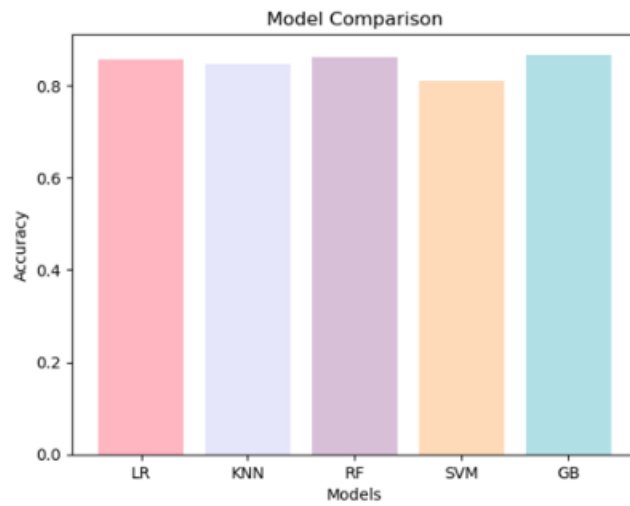
The models were evaluated using the following metrics:

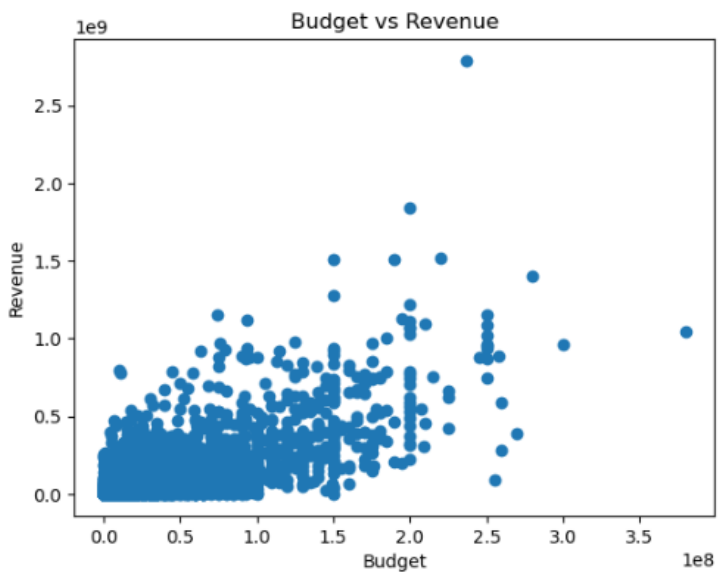
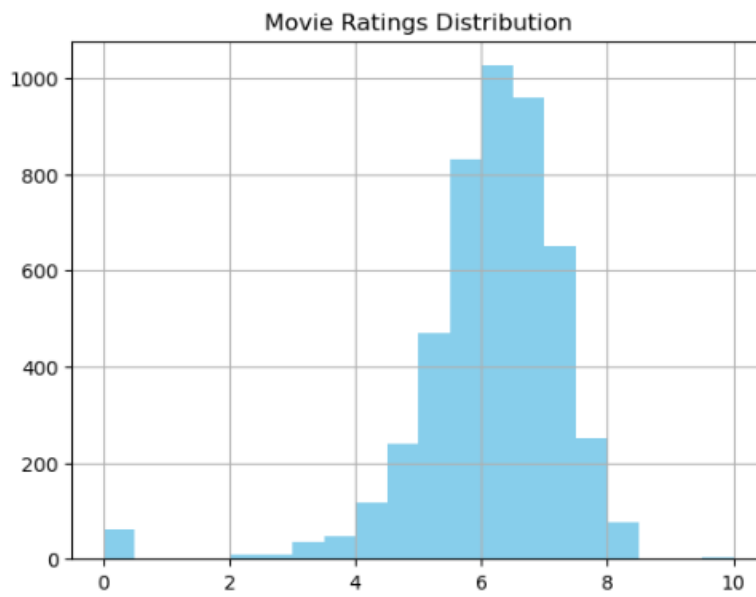
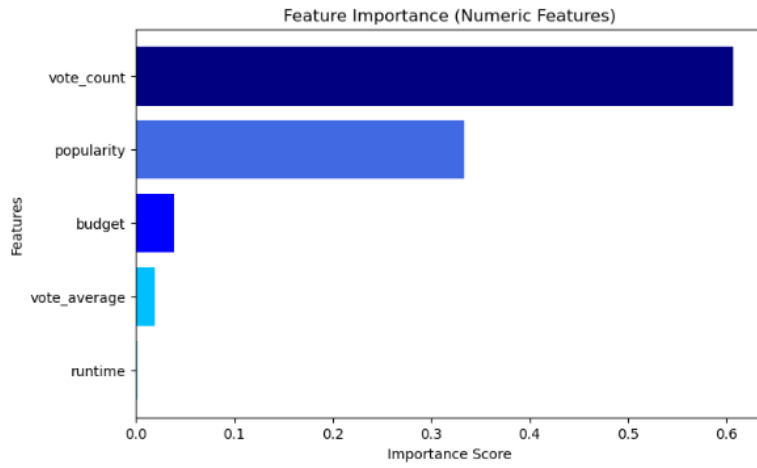
- Accuracy
- Precision
- Recall
- F1-score

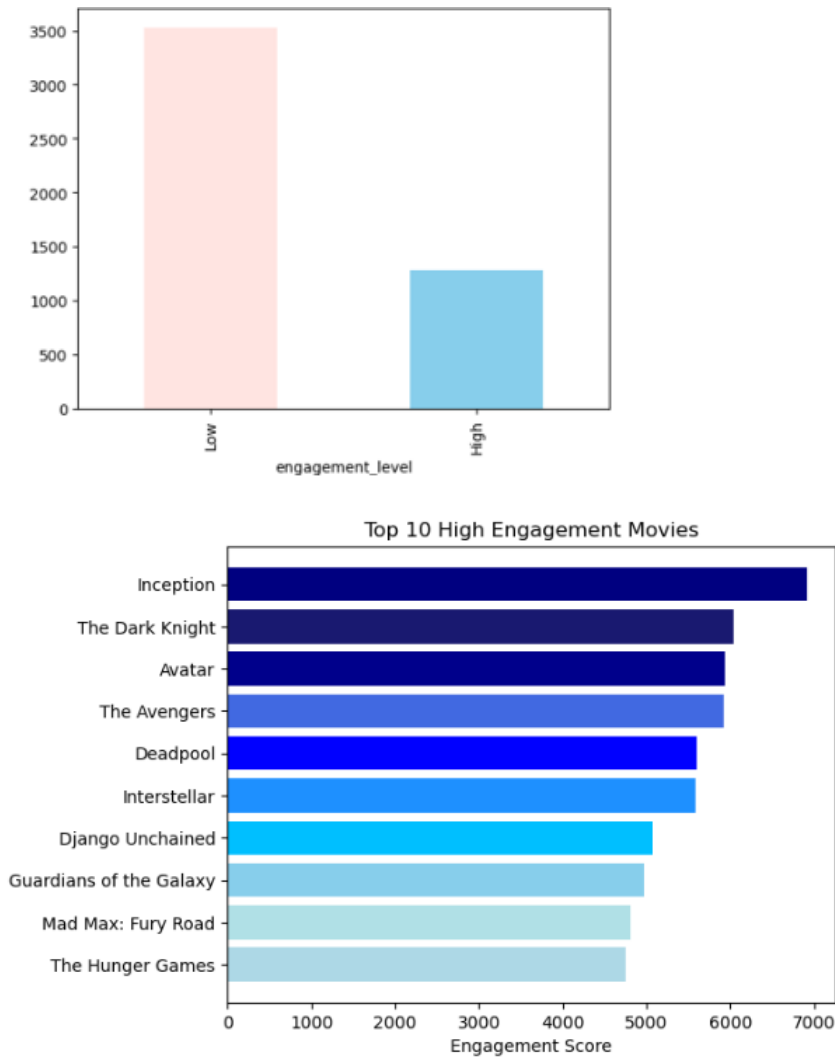
```
[75]: from sklearn.metrics import classification_report
      print(classification_report(y_test, y_pred_rf))
```

	precision	recall	f1-score	support
0	0.88	0.84	0.86	491
1	0.84	0.88	0.86	470
accuracy			0.86	961
macro avg	0.86	0.86	0.86	961
weighted avg	0.86	0.86	0.86	961

6.2. VISUALISATIONS







6.3. RESULT ANALYSIS

The analysis of model performance revealed that Random Forest achieved higher accuracy compared to other models, owing to its ensemble learning capability and ability to handle complex data patterns. Support Vector Machine (SVM) also delivered strong performance, effectively classifying data with clear decision boundaries.

On the other hand, Logistic Regression showed comparatively lower accuracy, as it is limited in capturing non-linear relationships within the dataset.

Overall, the results indicate that ensemble-based models are more effective for predicting audience engagement decline, as they provide better generalization and improved prediction accuracy.

7. DISCUSSION

The results of this study highlight the effectiveness of machine learning techniques in predicting audience engagement decline in movies and web series. The performance of different models shows that algorithms capable of handling complex and non-linear relationships tend to perform better.

Ensemble methods such as Random Forest and Gradient Boosting demonstrated superior performance due to their ability to combine multiple weak learners and reduce prediction errors. In contrast, simpler models like Logistic Regression were less effective, as they assume linear relationships between features.

K-Nearest Neighbors (KNN) provided moderate results, but its performance was influenced by factors such as feature scaling and the choice of the number of neighbors. Support Vector Machine (SVM) performed well in separating classes but may require careful parameter tuning for optimal results.

Overall, the study emphasizes the importance of data quality, proper preprocessing, and feature selection in improving model performance. While the models produced satisfactory results, their effectiveness can be further enhanced by incorporating more diverse and real-time data.

8. CONCLUSION AND FUTURE WORK

This project successfully demonstrates the application of machine learning techniques to predict audience engagement decline in movies and web series. By analyzing various features such as ratings, genre, and content patterns, the models were able to identify trends that contribute to declining engagement.

Among the implemented models, ensemble methods like Gradient Boosting and Random Forest showed the highest accuracy, making them the most suitable for this type of prediction problem. The results confirm that machine learning can provide valuable insights for content creators and streaming platforms to improve audience retention and content quality.

Future Work:

The project can be further enhanced in several ways:

- Incorporating real-time user interaction data from streaming platforms
- Applying deep learning techniques for improved accuracy
- Expanding the dataset to include more diverse and global content
- Developing a real-time prediction system or dashboard
- Integrating sentiment analysis from reviews and social media

These improvements can make the system more robust, scalable, and applicable in real-world scenarios.

9. REFERENCES

- Research papers from google and google scholar
- Kaggle datasets on movies and web series
- Scikit-learn documentation
- Various academic journals on machine learning applications
- ChatGPT and Claude ai for help